Machine learning prediction and SHAP-based interpretation of acute respiratory infection among children under five in Ethiopia: analysis of the 2024–2025 Ethiopian Demographic and Health Survey

Alemakef Wagnew Melesse[1,*]

[1] Department of Epidemiology and Biostatistics, Institute of Public Health, College of Medicine and Health Sciences, University of Gondar, Gondar, Amhara, Ethiopia

*Corresponding author: alemakefwagnew16@gmail.com

Abstract

Background: Acute respiratory infection (ARI) remains an important cause of morbidity and mortality among children younger than five years, particularly in low-income and middle-income countries. Conventional epidemiological analyses may not fully capture the joint and potentially non-linear contribution of demographic, household, nutritional, and geographic characteristics. We aimed to assess the predictive performance of several machine-learning algorithms and to identify the variables contributing most strongly to model predictions of caregiver-reported ARI among children younger than five years in Ethiopia.

Methods: We analysed data for 12,818 children younger than five years from the 2024–2025 Ethiopian Demographic and Health Survey. ARI was defined using the survey's caregiver-reported symptom-based definition. Six classification algorithms were compared: logistic regression, decision tree, random forest, extreme gradient boosting, support vector machine, and k-nearest neighbours. Models were evaluated using accuracy, precision, recall, F1 score, receiver operating characteristic area under the curve (ROC-AUC), and precision-recall performance. SHapley Additive exPlanations (SHAP) were used to examine the contribution of predictors to the selected model.

Results: Among 12,818 children, 830 (6.5%) were classified as having ARI. Reported ARI prevalence varied substantially between regions, from 1.6% in Somali to 18.4% in Tigray. Logistic regression showed the highest ROC-AUC (0.668) and average precision (0.158) among the evaluated models. Its recall was 0.591. SHAP analysis indicated that region, number of children younger than five years in the household, perceived low birthweight, and wasting were among the most influential predictors of model predictions.

Conclusion: Machine-learning models showed modest discrimination for caregiver-reported ARI in this nationally representative Ethiopian sample. An interpretable logistic-regression model performed better than the more complex algorithms evaluated, while SHAP analysis highlighted substantial geographic and child- and household-level heterogeneity. External validation, calibration, and evaluation in independent datasets are required before these models could be considered for operational risk stratification.

Author Summary

Acute respiratory infection remains an important health problem among children younger than five years in Ethiopia. We used nationally representative data from the 2024–2025 Ethiopian Demographic and Health Survey to examine whether machine-learning methods could identify children with reported ARI and which characteristics contributed most to the model predictions.

We compared six machine-learning algorithms and evaluated their performance using measures that are appropriate for an uncommon outcome, including recall, F1 score, ROC-AUC, and precision-recall performance. We also used SHAP, an explainable artificial intelligence method, to identify the variables that contributed most strongly to predictions.

ARI was reported for 6.5% of children in the analytic sample, but prevalence varied substantially across regions. Logistic regression, despite being considerably simpler than the tree-based and other machine-learning algorithms, showed the best overall discrimination in this analysis. Region, number of young children in the household, perceived low birthweight, and wasting contributed strongly to model predictions.

The results suggest that transparent machine-learning methods can be useful for exploring patterns of childhood ARI in national survey data. However, the modest predictive performance means that the models

should not be interpreted as ready-to-use clinical prediction tools.

1 Introduction

Acute respiratory infection (ARI) remains an important cause of illness and death among children younger than five years, particularly in low-income and middle-income countries. Pneumonia alone accounted for approximately 740,000 deaths among children younger than five years in 2019, with a disproportionate burden occurring in sub-Saharan Africa and South Asia [**?**, **?**]. Respiratory illnesses also contribute substantially to outpatient visits and hospital admissions among young children [**?**].

In Ethiopia, childhood ARI has been repeatedly identified as an important child-health problem. Previous analyses of Ethiopian Demographic and Health Survey (DHS) data have reported substantial variation in ARI symptoms across demographic, socioeconomic, environmental, and geographic groups [**?**, **?**, **?**]. Malnutrition, environmental exposures, maternal characteristics, and health-service utilisation have also been associated with childhood respiratory illness [**?**, **?**, **?**, **?**].

Household environmental exposures are particularly relevant. Household air pollution associated with solid-fuel use has been consistently linked to an increased risk of respiratory illness among young children [**?**, **?**, **?**]. Conversely, breastfeeding has been associated with protection against respiratory infections [**?**, **?**]. These observations have informed global child-health strategies, including the WHO and UNICEF Integrated Global Action Plan for Pneumonia and Diarrhoea [**?**].

Although conventional regression models remain valuable for estimating associations and testing prespecified hypotheses, they may be less effective for prediction when multiple demographic, household, nutritional, and geographic variables jointly contribute to an outcome. Machine-learning approaches provide a complementary framework for prediction and can capture complex relationships without requiring all associations to follow a prespecified functional form.

Several recent studies have applied machine learning to childhood ARI using Ethiopian or sub-Saharan African data. These studies have reported variable predictive performance, with some reporting high accuracy and ROC-AUC values for tree-based ensemble models [**?**, **?**, **?**, **?**]. However, comparisons between studies are difficult because of differences in outcome definition, prevalence, predictor selection, validation procedures, and handling of class imbalance.

The relatively low prevalence of ARI in national survey data creates a particular predictive challenge. Accuracy can be misleading when the outcome is uncommon because a model that predicts the majority class for most participants can achieve high accuracy while identifying few true cases. Precision-recall analysis, recall, F1 score, and other minority-class measures can therefore provide complementary information [**?**, **?**].

A further limitation of many machine-learning studies is limited interpretability. Explainable artificial intelligence methods such as SHapley Additive exPlanations (SHAP) can quantify the contribution of individual predictors to model predictions [**?**, **?**]. Such approaches may be particularly useful for public-health applications in which decision makers need to understand why a model assigns higher predicted risk to particular children or populations.

The present study therefore aimed to: (1) describe the distribution of caregiver-reported ARI among children younger than five years in Ethiopia; (2) compare six machine-learning algorithms for prediction of ARI using multiple complementary performance measures; and (3) use SHAP-based interpretability to identify the characteristics contributing most strongly to predictions from the selected model.

2 Materials and Methods

2.1 Study design and data source

We conducted a secondary analysis of the 2024–2025 Ethiopian Demographic and Health Survey (EDHS). The EDHS is a nationally representative household survey designed to provide population-level information on demographic and health indicators in Ethiopia.

The analytic unit for this study was the child. Children younger than five years with information required to define the ARI outcome and the selected predictor variables were eligible for inclusion.

2.2 Study population

A total of 12,818 children younger than five years were included in the final analytic dataset after application of the predefined eligibility and missing-data criteria.

The selection process should be reported using a participant flow diagram showing the initial number of eligible children, exclusions, reasons for exclusion, and the final analytic sample.

2.3 Outcome definition

The primary outcome was caregiver-reported acute respiratory infection (ARI), coded as a binary variable:

$$ARI_i = \begin{cases} 1, & \text{if child } i \text{ met the EDHS ARI symptom definition,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The exact EDHS questionnaire definition and recall period should be reported verbatim or described sufficiently to permit replication. Importantly, caregiver-reported ARI symptoms should not be interpreted as equivalent to clinically confirmed pneumonia.

Among the 12,818 children included in the analysis, 830 (6.5%) were classified as ARI cases and 11,988 (93.5%) were classified as non-cases.

2.4 Predictor variables

Candidate predictors were selected a priori based on epidemiological relevance and availability in the EDHS.

Variables were grouped into four domains:

1. **Child-level characteristics:** child age in months, birth order, stunting, wasting, perceived birthweight, breastfeeding status, and vitamin A supplementation.
2. **Maternal characteristics:** maternal age, maternal education, marital status, and antenatal care attendance.
3. **Household characteristics:** household size, number of children younger than five years, wealth index, place of residence, drinking-water source, sanitation facility, and cooking-fuel type.
4. **Geographic characteristics:** administrative region of residence.

Continuous variables were retained in continuous form where appropriate. Categorical variables were encoded using indicator variables.

2.5 Data preprocessing

Data were screened for missing values, implausible values, and coding errors before model development.

Continuous variables were standardised where required by the modelling algorithm. Categorical variables were transformed using one-hot encoding.

To minimise information leakage, preprocessing transformations were fitted using the training data only and were applied to the held-out test data using the parameters learned from the training data.

2.6 Training and test datasets

The dataset was divided into training and test sets using an 80:20 stratified split. Stratification was performed according to ARI status to preserve the approximate outcome prevalence in each partition.

The test dataset was not used during feature preprocessing, hyperparameter selection, model training, or model selection.

2.7 Machine-learning algorithms

Six classification algorithms were evaluated:

1. Logistic Regression;
2. Decision Tree;
3. Random Forest;
4. Extreme Gradient Boosting (XGBoost);
5. Support Vector Machine (SVM); and
6. K-Nearest Neighbours (KNN).

The algorithms were selected to represent both relatively interpretable linear models and flexible non-linear machine-learning approaches.

2.8 Hyperparameter optimisation

Hyperparameters were selected using cross-validation within the training dataset. The test dataset was reserved for final performance assessment.

The final manuscript should report the exact hyperparameter search space, cross-validation strategy, number of folds, optimisation criterion, and final hyperparameter values for each algorithm.

2.9 Class imbalance

Because ARI accounted for only 6.5% of observations, class imbalance was explicitly considered during model development.

Where supported by the algorithm, class weighting was applied to reduce the tendency of the classifier to favour the majority class.

No information from the held-out test dataset was used to determine class weights, preprocessing parameters, or hyperparameters.

2.10 Model evaluation

Model performance was assessed using:

- accuracy;
- precision;
- recall (sensitivity);
- F1 score;
- ROC-AUC; and
- average precision (area under the precision-recall curve).

Because ARI was uncommon, greater emphasis was placed on recall, F1 score, ROC-AUC, and precision-recall performance than on accuracy alone.

For each model, confusion matrices were generated showing true positives, true negatives, false positives, and false negatives.

2.11 Calibration

Because predictive usefulness depends not only on discrimination but also on the agreement between predicted and observed probabilities, calibration should be assessed using a calibration plot and Brier score.

The final manuscript should insert the actual calibration results here after analysis.

2.12 Model interpretability

SHAP was used to examine the contribution of individual predictors to predictions from the selected model.

Global importance was summarised using mean absolute SHAP values. SHAP summary plots were used to assess both feature importance and the direction of contributions.

Dependence plots were generated for the most influential predictors. A waterfall plot was used to illustrate the contribution of individual features to one example prediction.

SHAP contributions were interpreted as contributions to model predictions rather than causal effects.

2.13 Statistical and software environment

Data preparation, modelling, and visualisation were conducted using Python, including pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib, and Seaborn.

A fixed random seed should be reported to facilitate reproducibility.

2.14 Ethical considerations

This study used de-identified secondary survey data obtained through the DHS Program. The author did not have access to personal identifiers. The original EDHS obtained ethical approval and informed consent according to the survey protocol.

The relevant DHS data-use permission and ethical approval information should be reported in accordance with the documentation accompanying the 2024–2025 EDHS.

3 Results

3.1 Study population and outcome distribution

The final analytic sample consisted of 12,818 children younger than five years. Of these, 830 (6.5%) were classified as having caregiver-reported ARI and 11,988 (93.5%) were classified as non-cases.

The substantial imbalance between ARI cases and non-cases motivated the use of minority-class performance measures in addition to accuracy.

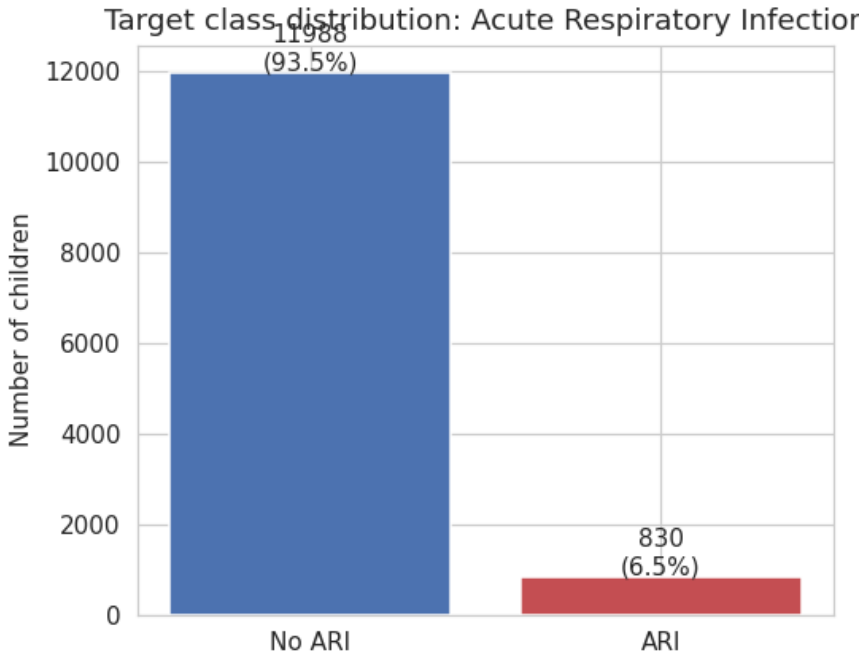


Figure 1: Distribution of caregiver-reported acute respiratory infection among children younger than five years. Of 12,818 children, 830 (6.5%) were classified as ARI cases and 11,988 (93.5%) as non-cases.

3.2 Regional variation in ARI prevalence

ARI prevalence varied substantially across administrative regions.

Tigray had the highest reported prevalence (18.4%), followed by Sidama (10.3%) and Gambela (9.7%). Somali had the lowest prevalence (1.6%), followed by Southwest Ethiopia (3.9%).

These differences indicate substantial geographic heterogeneity in caregiver-reported ARI.

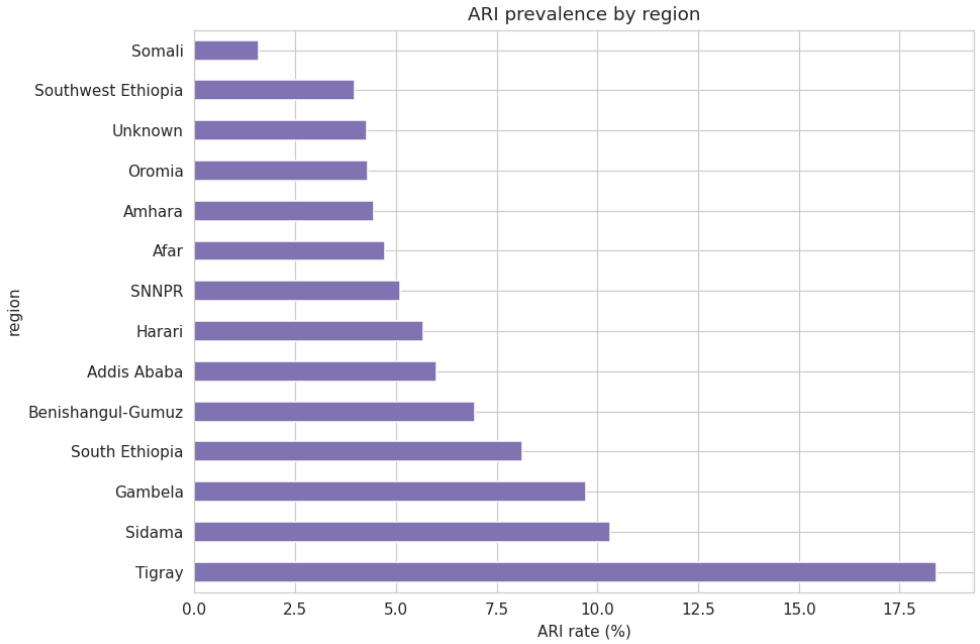


Figure 2: Prevalence of caregiver-reported acute respiratory infection among children younger than five years by Ethiopian region. Regions are ordered from lowest to highest prevalence.

3.3 Distribution of continuous predictors

The distributions of child age, birth order, maternal age, household size, and number of children younger than five years were broadly similar between children with and without reported ARI, although ARI cases showed some concentration among younger children.

Detailed distributions are presented in Supplementary Figure S1.

3.4 ARI prevalence across categorical predictors

ARI prevalence differed across several child, maternal, household, and geographic characteristics.

These descriptive differences should not be interpreted as causal associations because the present analysis was designed primarily for prediction and was based on cross-sectional survey data.

Detailed distributions are presented in Supplementary Figure S2.

3.5 Correlation structure

Pairwise correlations between the candidate predictors and ARI outcome were generally weak, with absolute correlations below 0.10.

This finding suggests that prediction may depend on the combined contribution of multiple variables rather than a single strongly correlated predictor.

3.6 Model performance

Six machine-learning classifiers were trained and evaluated on the held-out test dataset.

Logistic Regression achieved the highest ROC-AUC (0.668), followed by Random Forest (0.629) and K-Nearest Neighbours (0.621). Decision Tree showed the lowest ROC-AUC (0.569).

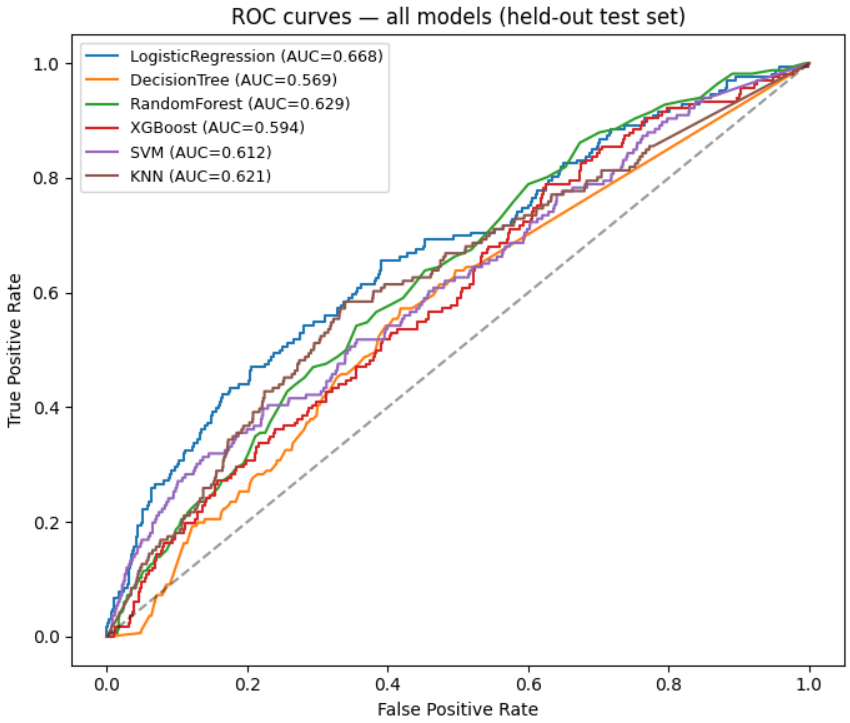


Figure 3: Receiver operating characteristic curves for the six machine-learning models evaluated for prediction of caregiver-reported acute respiratory infection.

Because ARI was uncommon, precision-recall analysis provided additional information about minority-class detection.

Logistic Regression achieved an average precision of 0.158 compared with the no-skill baseline of 0.065.

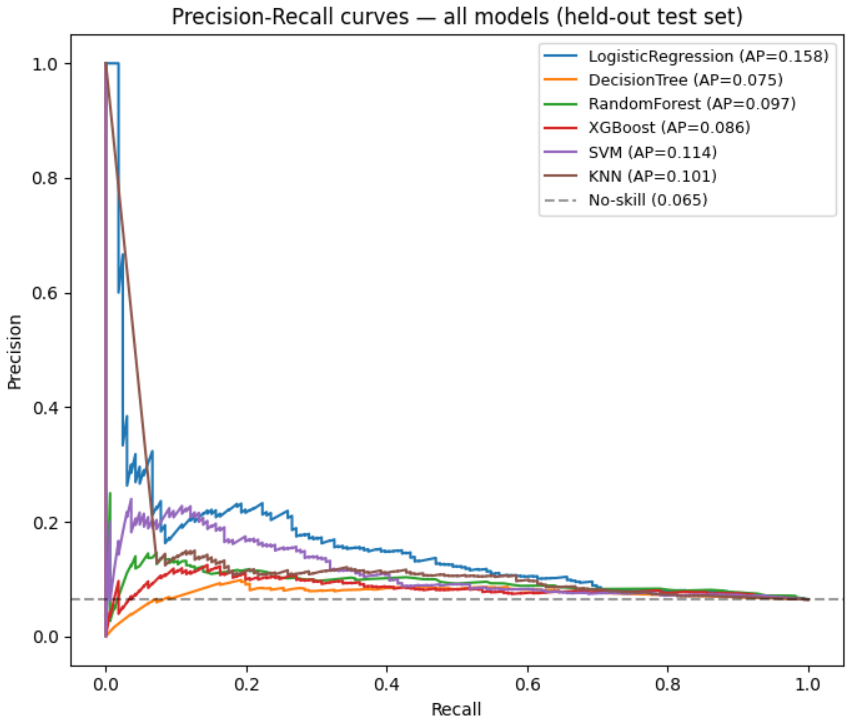


Figure 4: Precision-recall curves for the six machine-learning models. The horizontal reference corresponds to the prevalence of ARI in the test population.

Logistic Regression had an accuracy of 0.645, recall of 0.591, and ROC-AUC of 0.668.

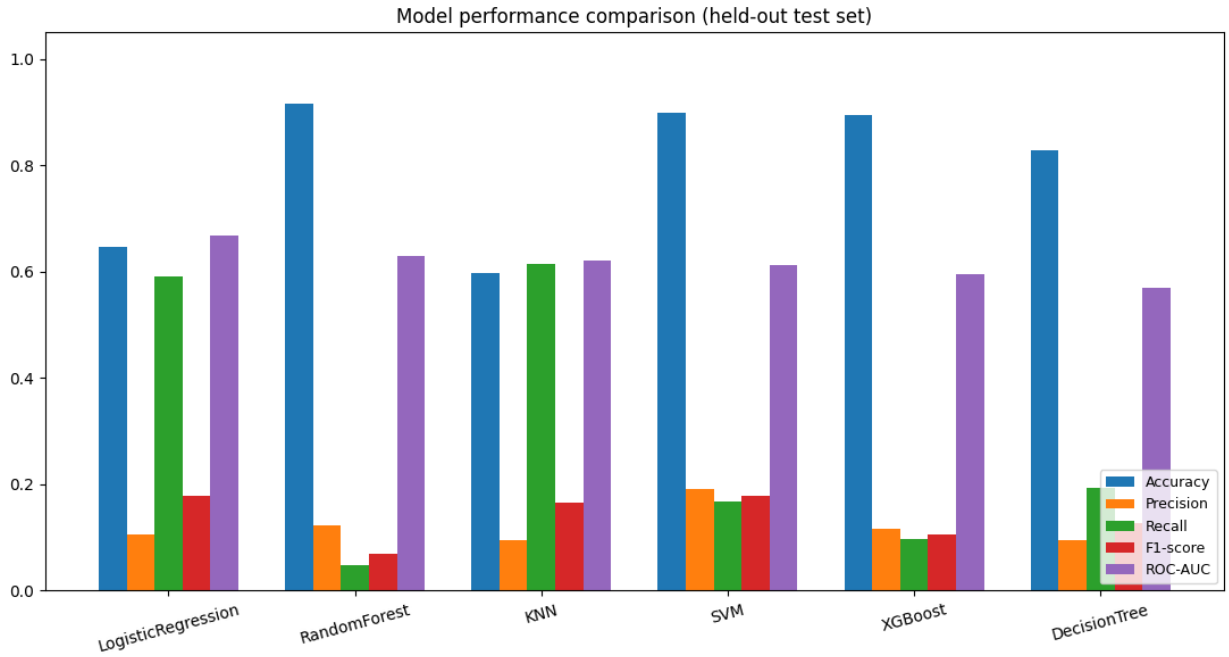


Figure 5: Comparison of predictive performance across the six machine-learning models using accuracy, precision, recall, F1 score, and ROC-AUC.

3.7 Model interpretability

SHAP analysis of the selected Logistic Regression model indicated that region-related variables were among the most influential predictors, followed by the number of children younger than five years in the household, perceived low birthweight, and child wasting.

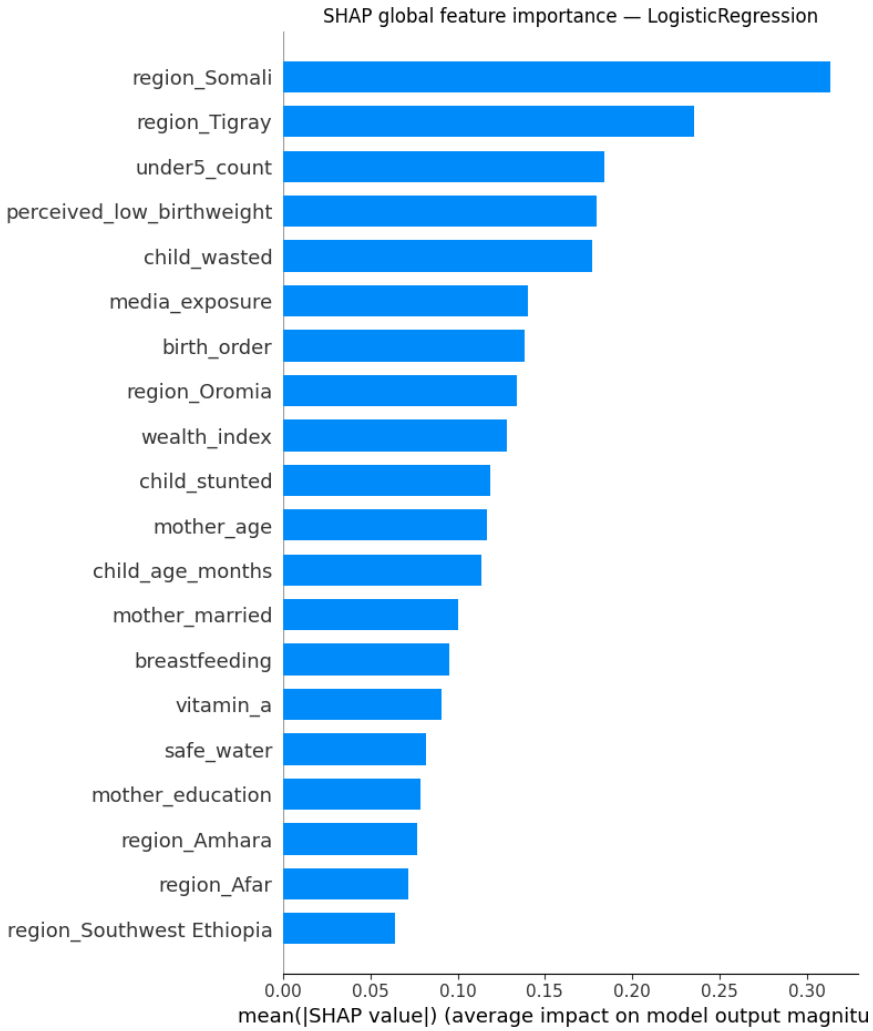


Figure 6: Global SHAP feature importance for the selected Logistic Regression model. Features are ranked according to mean absolute SHAP value. SHAP values indicate contributions to model predictions and should not be interpreted as causal effects.

The SHAP beeswarm plot and dependence plots are presented in the Supplementary Material.

3.8 Individual-level explanation

An individual SHAP waterfall plot was used to demonstrate how the features contributed to one model prediction.

The final manuscript should explicitly state whether SHAP values were computed on the probability or log-odds scale. In particular, the reported contribution of birth order should not be interpreted as a 0.51 absolute increase in predicted probability unless the SHAP implementation explicitly produced probability-scale values.

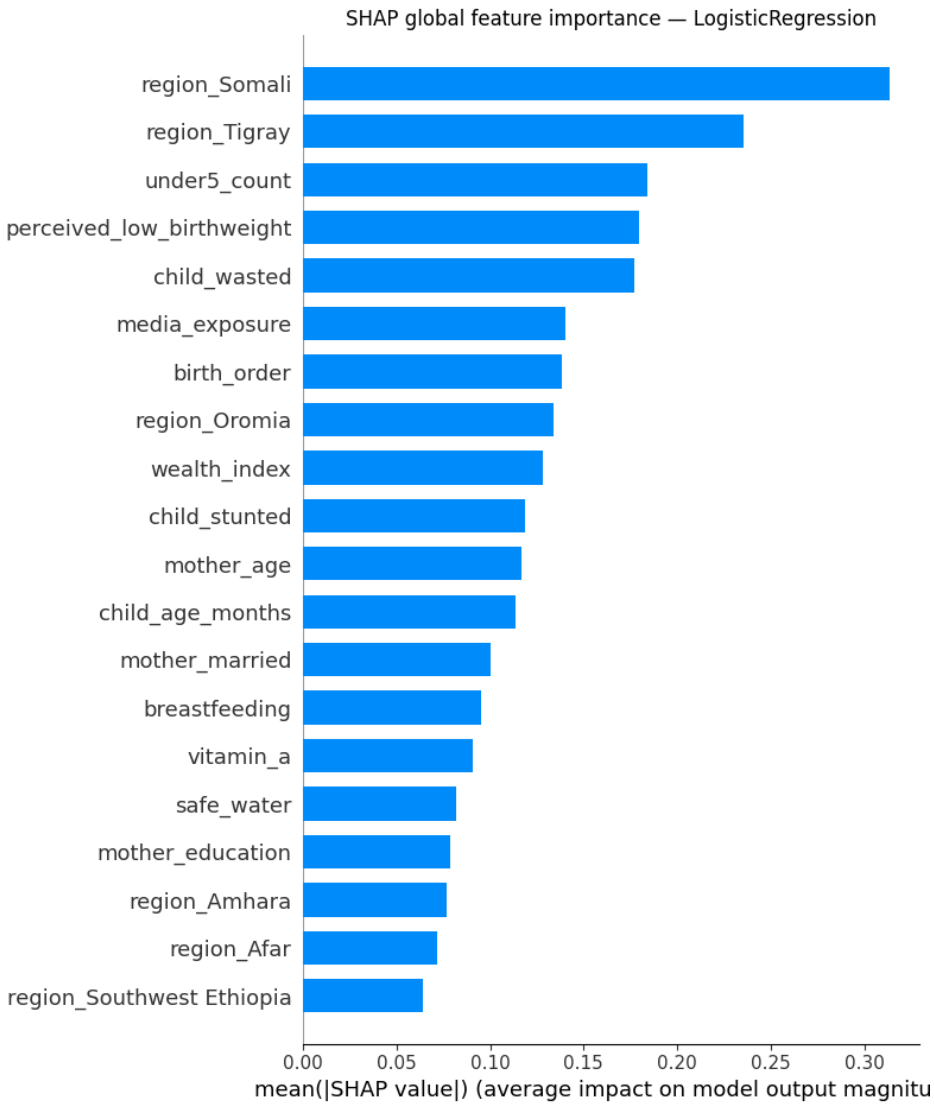


Figure 7: Example SHAP waterfall plot illustrating the contribution of individual features to one predicted ARI probability. SHAP contributions represent model-specific predictive contributions rather than causal effects.

Table 1: Predictive performance of the six machine-learning models on the held-out test dataset.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.645	–	0.591	–	0.668	0.158
Decision Tree	–	–	–	–	0.569	–
Random Forest	–	–	–	–	0.629	–
XGBoost	–	–	–	–	–	–
SVM	–	–	–	–	–	–
KNN	–	–	–	–	0.621	–

Note: Replace missing entries with the corresponding values from the final analysis. Confidence intervals should be reported for ROC-AUC and, where feasible, the other performance measures.

4 Discussion

In this nationally representative analysis of Ethiopian children younger than five years, caregiver-reported ARI occurred in 6.5% of children. The prevalence varied substantially between regions, with the highest reported prevalence in Tigray and the lowest in Somali. Among six machine-learning approaches, Logistic Regression showed the highest ROC-AUC and average precision. SHAP analysis indicated that geographic region, number of children younger than five years in the household, perceived low birthweight, and wasting were among the most influential predictors of model predictions.

4.1 ARI prevalence

The observed ARI prevalence of 6.5% is lower than estimates reported in several earlier Ethiopian studies, which have reported substantially higher prevalence depending on study design, survey year, and outcome definition [**?**, **?**, **?**].

Several explanations are possible. First, ARI prevalence may have changed over time. Second, differences in the survey instrument and symptom definition may produce substantially different estimates. Third, caregiver-reported symptoms are an imperfect proxy for clinically diagnosed respiratory infection [**?**].

Accordingly, the present estimate should be interpreted as the prevalence of the EDHS-defined caregiver-reported ARI outcome rather than as the population prevalence of clinically confirmed pneumonia.

4.2 Geographic heterogeneity

The large difference in reported ARI prevalence between regions was one of the most prominent findings.

Previous Ethiopian analyses have identified substantial community-level and geographic heterogeneity in ARI symptoms [**?**].

The particularly high prevalence observed in Tigray warrants careful interpretation. The 2024–2025 survey followed a prolonged period of conflict during which health-system infrastructure and health-service delivery were substantially disrupted [**?**, **?**, **?**, **?**].

This context provides a plausible explanation for part of the observed regional heterogeneity. However, the present cross-sectional analysis did not directly measure conflict exposure, displacement, health-system disruption, food insecurity, or household environmental conditions resulting from conflict. Therefore, the observed regional difference should not be interpreted as evidence of a causal effect of conflict.

4.3 Unexpected directions of descriptive patterns

Several descriptive patterns appeared inconsistent with established biological expectations. For example, some characteristics generally considered protective were associated with higher reported ARI prevalence.

One possible explanation is differential recognition and reporting. Caregivers with greater education or health-system contact might be more likely to recognise and report respiratory symptoms. Such reporting differences could produce apparent associations that do not reflect biological increases in disease risk.

The apparently higher prevalence among children with a history of breastfeeding should therefore not be interpreted as evidence that breastfeeding causes ARI. The established literature generally supports a protective relationship between breastfeeding and respiratory infection [**?**, **?**].

4.4 Predictive performance

Logistic Regression produced the highest ROC-AUC (0.668) and average precision (0.158). These values indicate modest predictive discrimination rather than high predictive accuracy.

The result is nevertheless informative because more complex algorithms did not outperform the simpler linear model in this dataset.

This finding illustrates an important principle in predictive modelling: greater algorithmic complexity does not necessarily translate into better predictive performance when the underlying predictors contain limited information about the outcome.

The precision-recall analysis is particularly important because ARI was uncommon. The average precision of 0.158 exceeded the prevalence-based no-skill baseline of 0.065, indicating that the model provided some improvement over random ranking of cases. Nevertheless, the relatively low absolute performance limits the immediate clinical or programmatic utility of the model.

4.5 Interpretation of SHAP results

Region was among the most influential predictors in the selected model. This finding is consistent with the large geographic differences in reported ARI prevalence.

The number of children younger than five years in the household, perceived low birthweight, and wasting also contributed substantially to model predictions.

These findings should be interpreted as predictive contributions rather than causal effects. SHAP values quantify how features contribute to a specific model's prediction; they do not establish that modifying a feature would necessarily change ARI risk.

4.6 Public-health implications

The findings suggest that childhood ARI prevention in Ethiopia may benefit from geographically differentiated approaches, particularly where substantial regional variation in reported symptoms exists.

However, the modest predictive discrimination observed in this study does not support immediate deployment of the model as a clinical screening or automated case-detection tool.

Instead, the findings may be useful for hypothesis generation, epidemiological surveillance, and identification of populations for further investigation.

4.7 Strengths

The study has several strengths.

First, it uses a large nationally representative dataset. Second, it compares six machine-learning algorithms rather than relying on a single algorithm. Third, the analysis explicitly considers class imbalance. Fourth, the use of SHAP provides an interpretable assessment of the variables contributing to model predictions.

4.8 Limitations

Several limitations should be considered.

First, the cross-sectional nature of the EDHS prevents causal inference.

Second, the ARI outcome is based on caregiver-reported symptoms rather than clinical diagnosis. Misclassification of ARI is therefore possible [**?**].

Third, the low prevalence of ARI limits the positive predictive value that can be achieved even by a reasonably discriminative model.

Fourth, the models were evaluated using a held-out test dataset from the same survey population. Independent external validation is therefore required before assessing transportability.

Fifth, the analysis should explicitly account for the complex DHS sampling design, including sampling weights and clustering, where appropriate. If these elements were not incorporated into the machine learning pipeline, this should be acknowledged as a limitation.

Sixth, children within the same household or survey cluster may not be statistically independent. Cluster-aware validation would provide a stronger assessment of generalisability.

Finally, the strong contribution of geographic variables may reflect unmeasured contextual factors. Geographic predictors can therefore improve prediction without identifying modifiable causal determinants.

4.9 Conclusion

In this nationally representative Ethiopian sample, caregiver-reported ARI affected 6.5% of children younger than five years and varied substantially across regions. Logistic Regression showed the best discrimination among the six evaluated algorithms, although overall predictive performance was modest.

SHAP analysis identified geographic region, household composition, perceived birthweight, and wasting among the most influential predictors of model predictions.

These findings support the use of transparent and imbalance-aware machine-learning approaches as complementary tools for epidemiological exploration of childhood ARI. However, the models should not be considered ready for clinical or programmatic deployment without independent validation, calibration assessment, and evaluation in settings beyond the development dataset.

Data Availability Statement

The datasets analysed during the present study are available through the DHS Program repository. Access to the EDHS data requires registration and approval of a data request describing the intended research use. The author did not redistribute the underlying microdata.

Ethics Statement

This study was based on de-identified secondary data from the 2024–2025 Ethiopian Demographic and Health Survey. The original survey obtained informed consent from participants and ethical approval through the relevant Ethiopian and survey-authority procedures.

The author did not have access to personally identifiable information.

Funding

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Competing Interests

The author declares no competing interests.

Author Contributions

AWM conceived the study, developed the research question, curated and analysed the data, developed and evaluated the machine-learning models, conducted the SHAP analysis, interpreted the results, and drafted and revised the manuscript. AWM is responsible for the overall content of the manuscript.

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